

Sukumar Kavishan

Software Engineer

Colombo, Sri Lanka | kavishansukumar@gmail.com | +94 77 584 4312 | linkedin.com/in/sukumar-kavishan
github.com/KavishanSukumar

Professional Summary

Specializing in mission-critical **financial systems**, including capital markets, stock trading platforms, and government securities management. Strong backend engineering expertise (Java, Quarkus, Spring Boot) coupled with a dedicated focus on **application security** (Keycloak, OAuth2, RBAC, automated secure code review). Proficient in leveraging **AI-assisted engineering workflows** (Azure DevOps, Gemini MCP) to accelerate secure software delivery. Deep understanding of the trade lifecycle, **financial instruments** (Equities, T-Bills, T-Bonds), and KYC/AML regulatory compliance in high-stakes environments.

Experience

Software Engineer, IronOne Technologies (Pvt) Ltd. (*Fintech – Capital Markets*) June 2024 - Present

Atrads Sri Lanka – Stock Market Order Management System

- Migrated market data delivery from polling to **WebSocket-based streaming**, minimizing latency for live order books and trade confirmations.
- Implemented **Redis distributed caching** for all market data retrieval, decreasing direct database hits by **72%** during peak trading hours.
- Resolved thread contention and deadlocks by integrating thread-safe data structures, dropping server CPU usage to **20%**.
- Integrated **ActiveMQ** to decouple order ingestion from downstream processing utilizing a robust **Publish-Subscribe** pattern.
- *Tech & Domain*: Equity Trading (CSE), Java, WebSocket, ActiveMQ, Redis, Multithreading, Performance Profiling.

Primary Dealer Management Platform – Government Securities

- Engineered core platform features to manage T-Bill/T-Bond auctions, Repurchase Agreements, and secondary market trades for licensed primary dealers.
- Enforced strict security and regulatory standards by implementing **Keycloak RBAC**, KYC workflow management, and CBSL-compliant audit trail logging.
- Containerized microservices with **Docker/Kubernetes** on AWS and automated database schema migrations using **Flyway**.
- Integrated Azure DevOps with **Gemini MCP** for AI-assisted PR reviews, automating secure-coding and vulnerability checks to improve code quality and turnaround time.
- *Tech & Domain*: Fixed Income, Regulatory Compliance, Quarkus, React (TS), Keycloak, Flyway, Docker, K8s, AWS, Gemini MCP.

Wealth Management System

- Engineered End-of-Day (EOD) processing pipelines for cross-instrument portfolio reconciliation and developed comprehensive multi-asset reporting modules.
- Implemented automated **SWIFT message generation** for DVP/RVP instructions, enabling straight-through securities settlement with custodian banks.
- Extended AI-assisted PR review automation to enforce secure-coding practices and consistency across portfolio and transaction modules.
- *Tech & Domain*: Multi-Asset Portfolio Management, SWIFT Messaging, Quarkus (Reactive), React (TS), Docker, K8s, AWS.

Software Engineer Intern, Inova IT Systems (Pvt) Ltd.

November 2022 - May 2023

- **ClickUp Dashboard**: Developed a scalable dynamic project analytics dashboard utilizing React (TS), Spring Boot, and Power BI for advanced data visualization.
- **CRM Mobile Backend**: Designed resilient Spring Boot backend REST APIs to streamline internal ticket issuance and allocation workflows.

Freelance Software Developer (*Part-Time*)

January 2021 - June 2024

- Developed and delivered monolithic PHP/MySQL web applications for local clients, independently managing the full software lifecycle from requirements analysis to deployment.

Financial Domain Knowledge

- **Capital Markets:** Equity trading lifecycle (order placement, matching, settlement), real-time market data feeds, order book management, CSE operations.
- **Fixed Income & Government Securities:** T-Bills, T-Bonds, primary/secondary market auction processes, yield and pricing concepts, primary dealer obligations under CBSL guidelines.
- **Regulatory & Compliance:** KYC/AML workflows, role-based access control for financial platforms, audit trail requirements, regulatory reporting patterns, CBSL compliance considerations.
- **Fintech Architecture:** High-throughput transaction processing, event-driven financial data pipelines, secure API design for financial services, idempotent transaction handling, distributed caching.

Technical Skills

Programming Languages: Java, Python, JavaScript, TypeScript, C++ , Dart, C
Frameworks & Libraries: Spring Boot, Quarkus, React, React Native, Flutter, Node.js, Express.js
Databases & Data Stores: PostgreSQL, MySQL, MongoDB, Redis
Messaging & Streaming: ActiveMQ, WebSocket, UDP, Event-driven Architecture
Cloud, DevOps & Tooling: AWS, Docker, Kubernetes, Git, Linux, Keycloak, Flyway, Shell Scripting, Azure DevOps, CI/CD
Application Security: Secure API Design, OWASP Top 10, Secure Code Review, RBAC, OAuth2, Automated Security Checks

Projects

- Nocturne – Real-Time Driving Assistance
4th Year Research Group Project, UCSC

YOLOv8, PyTorch, React, Spring Boot

 - Researched and integrated a **YOLOv8 object detection model** with live camera feeds to provide real-time night driving assistance.
 - Conducted extensive data collection and annotation to accurately detect potholes, cyclists, and pedestrians in low-light conditions.
- Entero – Event Management System
3rd Year Group Project, UCSC

Node.js, Express, React, PostgreSQL

 - Engineered core backend services including real-time chat, notifications, and robust authentication/authorization workflows.
 - Integrated Azure Blob Storage for scalable asset management within a monolithic application architecture.
- Construct360⁰ – Construction Management System
2nd Year Group Project, UCSC

Java, HTML, CSS, JavaScript

 - Developed critical platform features including real-time chat, secure payment gateway integration, and role-based access control modules.

ML & AI Research

- **Research Interests:** Natural Language Processing (NLP), Computer Vision, Real-time AI systems.
- **Publications:** Abstract published at the *Annual Research Symposium 2023*, University of Colombo. Accepted for presentation at *IECON 2024 – IEEE Industrial Electronics Society*.

Education

University of Colombo School of Computing (UCSC)
B.Sc. (Hons) in Software Engineering – Second Class Upper Division (GPA: 3.57 / 4.0)

March 2020 - June 2024